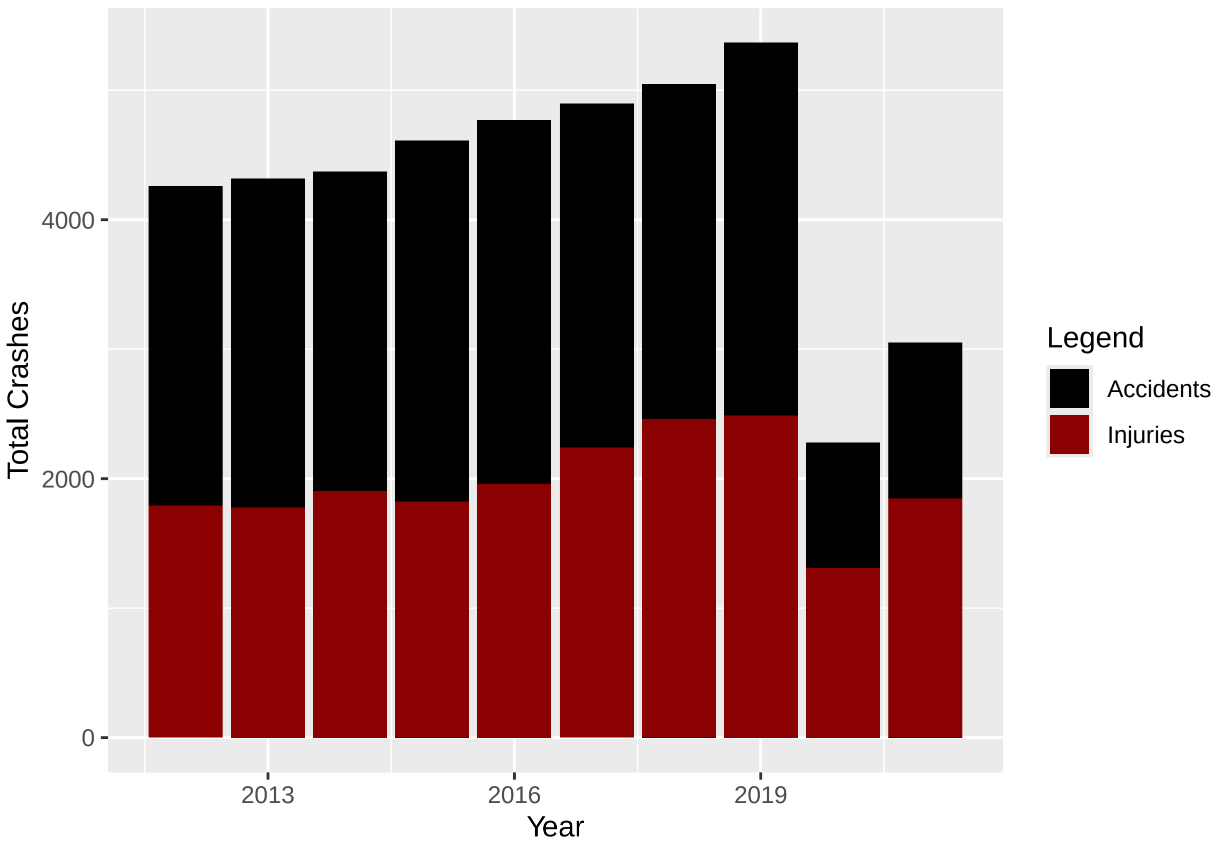
Final Portfolio

Shanna Barrett



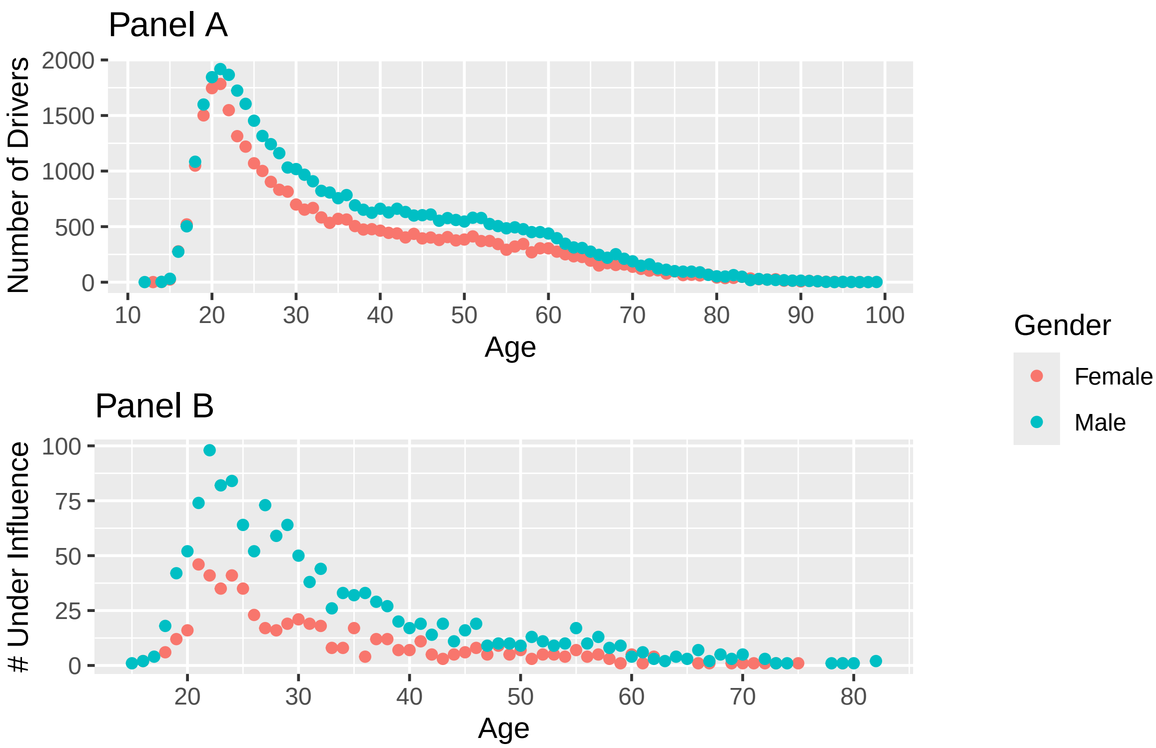
Plot 1 - This plot shows the total number of accidents and injuries in Tempe, Arizona from 2012-2021. With total number of accidents, in 2012-2014 there was a slow steady increase in accidents. There was a small jump from 2014-2015 and 2015-2018 there was a steady increase in accidents with another jump in 2019. Then there is a large decrease in the number of accidents in 2020. From 2020-2021, there is an about 700 increase in the number of accidents. The number of accidents in 2020 was about half as many as there was in 2019. The number of injuries roughly follows the trends of the accidents. One thing that differs is there was a decrease in the number of accidents in 2015 compared to 2014. The number of injuries compared to noninjuries is a larger proportion in accidents in 2020.

Alcohol Use	Drug Use	Total	Percent
Alcohol	Drugs	92	0.12%
Alcohol	No Apparent Influence	1990	2.59%
No Apparent Influence	Drugs	303	0.39%
No Apparent Influence	No Apparent Influence	74437	96.90%

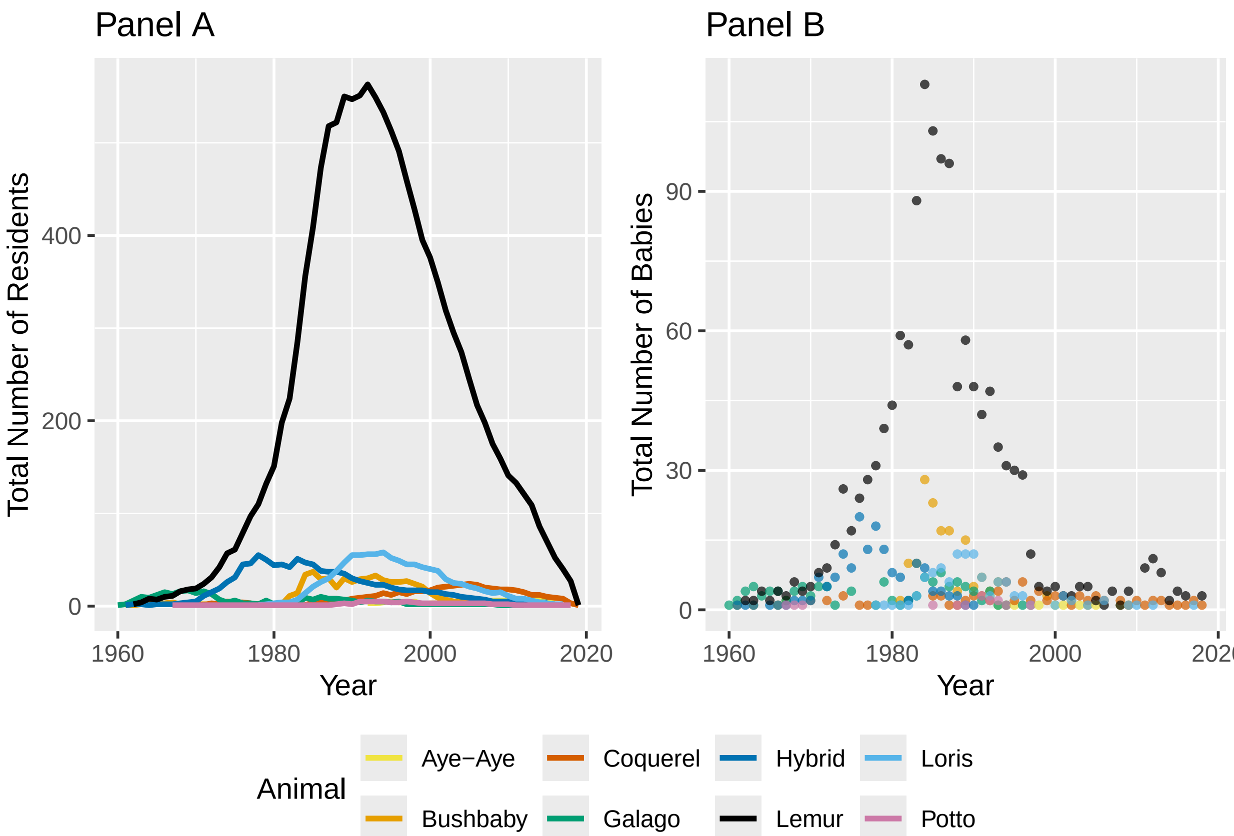
Table 1 - The table above shows the number of people involved in accidents in Tempe, Arizona from 2012-2021 that were on drugs, alcohol, both, or none. Most of the people involved in an accident were not under any sort of influence. Only about 3% of people in the dataset were under one or more substance.

Table Notes:

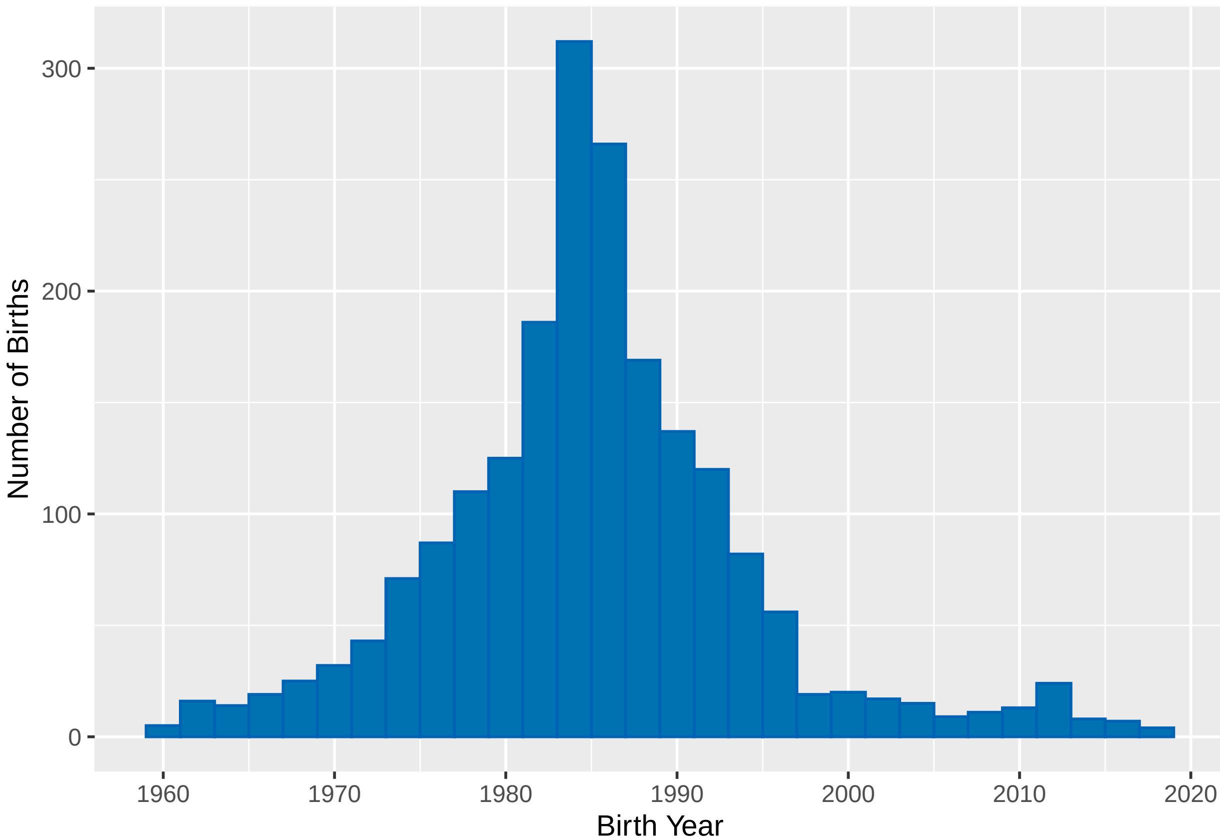
- The table reflects the number of people who fit into those categories
- It does not reflect the number of accidents



Plot 3 - Panel A is showing the gender and number of drivers involved in an accident in Tempe from 2012-2021 who are the ages from 15 to 100. The data was filtered where the Unit Type (Driver, Pedestrian, or Pedalcyclist) was only Drivers. Due to it being the age of drivers, not everyone involved in an accident, the ages start at 15 as that is the learners permit age in Arizona. Females, from the ages, 19 to around 80 were less likely to be seen in an accident in Tempe. Panel B is showing the ages of drivers who were under the influence of alcohol while driving. The youngest male driver was 15 while the oldest male driver was 82. Females overall were involved in less accidents and were less likely to drive while under the influence of alcohol. The figures answered the question of “How do males and females differ in the number of accidents? How does gender and substance use compare?”



Plot 4 - The figures above answered the questions, "How has the number of total residents at the Duke Lemur Center changed over time by primate type? How has the number of babies born at the Duke Lemur Center changed over time by primate type?" The figures above show the total number of residents present and the total number of babies born each year by primate type at the Duke Lemur Center. Panel A shows that overall, Lemurs have had the most yearly residents and had a max in the early 1990s. Panel B shows that yearly, lemurs have had the most babies and had a max in the mid 1980s. One thing to note is that Panel A and B have different scales on the y-axis. Panel A has a max of 600 while Panel B has a max of 120. All the colors correspond with the same animal on both plots.



Plot 5 - This histogram shows the distribution of animals being born at the Duke Lemur Center from 1960 to 2018. There was a peak in the numbers in 1983-1985 with over 300 births. After the late 1990s, the number of babies born stays consistent with numbers below 25. This plot answers the question of, “how does the total number of births change over time?”

Code Output

```r
Crash_Data <- read.csv("~/Downloads/Crash_Data_Report_(detail).csv")

library(dplyr)
library(ggplot2)
library(reshape2)
library(lubridate)
library(tidyr)
library(kableExtra)
library(gridExtra)

Per_Year <- Crash_Data %>%
  group_by(Year) %>%
  tally()

Per_Year <- Per_Year %>%
  rename(TotalAccidents = n)

#ggplot(Per_Year, aes(x = Year, y = TotalAccidents))+
#  geom_bar(stat = "identity", fill = "darkred")+
#  labs(x = "Year", y = "Number of Accidents")+
#  ylim(0,6000)

Just_Injuries <- Crash_Data %>%
  filter(Totalinjuries != 0)

Just_Injuries <- Just_Injuries %>%
  group_by(Year)%>%
  summarize(Injuries = sum(Totalinjuries))

ggplot()+
  geom_bar(stat = "identity", data = Per_Year, aes(x = Year, y = TotalAccidents,
           fill = "Accidents"), size = 1)+
  geom_bar(stat = "identity", data = Just_Injuries,
          aes(Year, Injuries, fill = "Injuries"), size = 1)+
  labs(x = "Year", y = "Total Crashes", fill = "Legend")+
  scale_fill_manual(values = c("black", "darkred"))
```

```r
library(scales)

Personal_info <- Crash_Data %>%
  select(Year, Age_Drv1, Gender_Drv1, Unittype_One, Violation1_Drv1, AlcoholUse_Drv1,
        DrugUse_Drv1, Age_Drv2, Gender_Drv2, Unittype_Two,Violation1_Drv2,
        AlcoholUse_Drv2, DrugUse_Drv2)

Driver1 <- Personal_info %>%
  select(Year, Age_Drv1, Gender_Drv1, Unittype_One, Violation1_Drv1, AlcoholUse_Drv1,
        DrugUse_Drv1)

Driver2 <- Personal_info %>%
  select(Year, Age_Drv2, Gender_Drv2, Unittype_Two, Violation1_Drv2, AlcoholUse_Drv2,
        DrugUse_Drv2)
```

```r
Driver1 <- Driver1 %>%
  rename(Age = Age_Drv1,
         Gender = Gender_Drv1,
         UnitType = Unittype_One,
         Violation = Violation1_Drv1,
         AlcoholUse = AlcoholUse_Drv1,
         DrugUse = DrugUse_Drv1)

Driver2 <- Driver2 %>%
  rename(Age = Age_Drv2,
         Gender = Gender_Drv2,
         UnitType = Unittype_Two,
         Violation = Violation1_Drv2,
         AlcoholUse = AlcoholUse_Drv2,
         DrugUse = DrugUse_Drv2)

Drivers = rbind(Driver1, Driver2)

#We are filtering out where Gender is unknown, and ages are not living ages.
#There is no note on what these numbers mean
#otherwise so we are filtering them out. They were all drivers as well so you
#can't assume it was driver 1 and an 11 year old.
#There wouldn't be that many 11 year old drivers
Drivers <- Drivers %>% filter(Gender != "", Gender != "Unknown", Age != 255,
                              Age != 113, Age != 114, Age != 115, Age != 116,
                              Age != 117, Age != 118, Age != 119, Age != 120,
                              Age != 111, Age != 112, Age != 254)

tally_alco <- Drivers %>%
  group_by(AlcoholUse, DrugUse)%>%
  tally()

tally_alco <- tally_alco %>%
  rename(Total = n)

tally_alco <- tally_alco %>%
  mutate(Percent = (Total/76822))

#tally_alco <- tally_alco %>%
 # mutate(across('Percent', round, 2))

#tally_alco <- tally_alco %>%
 # mutate(percent('Percent', accuracy = 0.1))


tally_alco$Percent <- percent(tally_alco$Percent, accuracy=0.01)



#This is the percentage of total people involved in accidents that
#were on drugs or alcohol or none
```

```r
kable(tally_alco, col.names = c("Alcohol Use", "Drug Use", "Total", "Percent"))
```

```r
Age_Drivers <- Drivers %>%
  group_by( Gender, Age) %>%
  filter(UnitType == "Driver")%>%
  tally()

A1 <- ggplot(Age_Drivers, aes(Age, n, col = Gender))+
  geom_point(stat = "identity")+
  xlim(0,100)+
  scale_x_continuous(breaks = seq(0, 100, by = 10))+
  labs(x = "Age",  y = "Number of Drivers", title = "Panel A")+
  theme(legend.position = "none")


#This plot is showing the gender and number of drivers involved in an accident
#in Tempe from 2012-2021 who are #the ages from 15 to 100. Due to it being the
#age of drivers, not everyone involved in an accident, the ages #start at 15 as
#that is the learners permit age in Arizona.

influenced_driver <- Drivers %>%
  filter(AlcoholUse == "Alcohol")

influenced_driver <- influenced_driver %>%
  group_by( Gender, Age, AlcoholUse) %>%
  filter(UnitType == "Driver")%>%
  tally()

A2 <- ggplot(influenced_driver, aes(Age, n, col = Gender))+
  geom_point(stat = "identity")+
  xlim(0,100)+
  scale_x_continuous(breaks = seq(0, 100, by = 10))+
  labs(x = "Age", y = "# Under Influence", title = "Panel B")+
  theme(legend.position = "none")


A3 <- grid.arrange(A1, A2, nrow = 2)
```

```r
plot2_legend <- ggplot(Age_Drivers, aes(Age, n, col = Gender))+
  geom_point(stat = "identity")+
  xlim(0,100)+
  scale_x_continuous(breaks = seq(0, 100, by = 10))+
  labs(x = "Age",  y = "Number of Drivers", title = "Panel A")+
  theme(legend.position = "right")

get_only_legend <- function(plot) {

# get tabular interpretation of plot
plot_table <- ggplot_gtable(ggplot_build(plot))

# Mark only legend in plot
legend_plot <- which(sapply(plot_table$grobs, function(x) x$name) == "guide-box")

# extract legend
```

```r
legend <- plot_table$grobs[[legend_plot]]

# return legend
return(legend)
}
legend <- get_only_legend(plot2_legend)

grid.arrange(A3, legend, nrow = 2, widths = c(9,3), heights = c(15,2))
```

```r
grid.arrange(A3, legend, nrow = 2, widths = c(9,3), heights = c(15,2))
```

```r
library(readxl)
DLCLH <- read_excel("~/Downloads/Duke Lemur Center Life History.xlsx")

library(dplyr)
library(ggplot2)
library(reshape2)
library(lubridate)
library(tidyr)
library(patchwork)

#this is filtering for only born at duke lemur
Born_Duke <- DLCLH %>%
  filter(Birth_Institution == 'Duke Lemur Center')

#This is seperating out years
Born_Duke <- Born_Duke %>%
  mutate(birthdate = as.POSIXct(DOB, format = "%Y-%m-%d")) %>%
  mutate(Year_Birth = year(birthdate)) %>%
  mutate(Month_Birth = month(birthdate)) %>%
  mutate(Day_Birth = day(birthdate))

#ditto
Born_Duke <- Born_Duke %>%
  mutate(deathdate = as.POSIXct(DOD, format = "%Y-%m-%d")) %>%
  mutate(Year_Death = year(deathdate)) %>%
  mutate(Month_Death = month(deathdate)) %>%
  mutate(Day_Death = day(deathdate))

#this is to check for na
#unique(Born_Duke$Year_Birth)
#unique(Born_Duke$Year_Death)

#filtering out the na
Born_Duke <- Born_Duke %>% filter(!is.na(Year_Birth))
Born_Duke <- Born_Duke %>% filter(!is.na(Year_Death))

#making sure that worked
#unique(Born_Duke$Year_Birth)
#unique(Born_Duke$Year_Death)

#This code is writing out and seperating taxon into specific groups
mutated_born <- Born_Duke %>%
  mutate(Animal = case_when(Taxon == "OGG" ~ "Galago",
```

```r
                            Taxon == "ECOL" ~ "Lemur",
                            Taxon == "EUL" ~ "Hybrid",
                            Taxon == "EFUL" ~ "Lemur",
                            Taxon == "ERUF" ~ "Lemur",
                            Taxon == "HGG" ~ "Lemur",
                            Taxon == "VRUB" ~ "Lemur",
                            Taxon == "CMED" ~ "Lemur",
                            Taxon == "PPOT" ~ "Potto",
                            Taxon == "GMOH" ~ "Bushbaby",
                            Taxon == "EMON" ~ "Lemur",
                            Taxon == "VVV" ~ "Lemur",
                            Taxon == "EMAC" ~ "Lemur",
                            Taxon == "LCAT" ~ "Lemur",
                            Taxon == "EALB" ~ "Lemur",
                            Taxon == "VAR" ~ "Hybrid",
                            Taxon == "PCOQ" ~ "Coquerel",
                            Taxon == "MMUR" ~ "Lemur",
                            Taxon == "LTAR" ~ "Loris",
                            Taxon == "NCOU" ~ "Loris",
                            Taxon == "NPYG" ~ "Loris",
                            Taxon == "MZAZ" ~ "Lemur",
                            Taxon == "ECOR" ~ "Lemur",
                            Taxon == "ERUB" ~ "Lemur",
                            Taxon == "ESAN" ~ "Lemur",
                            Taxon == "EFLA" ~ "Lemur",
                            Taxon == "DMAD" ~ "Aye-Aye"
  ))

#This is making a new dataframe with new birth years and animal and tallying those
Birth_duke <- mutated_born %>%
  group_by(Year_Birth, Animal)%>%
  tally()

Birth_duke <- Birth_duke %>% filter(!is.na(Year_Birth))

#unique(Birth_duke$Animal)
```

```r
#this is the beginning of the present year lemur count
Dif_Years <- mutated_born %>%
  select(Animal, Year_Birth, Year_Death)

y = data.frame(yearpres=0)

for( i in 1:length(Dif_Years$Year_Birth)){
  x = data.frame(yearpres = rep(Dif_Years$Year_Birth[i]:Dif_Years$Year_Death[i]))
  y = rbind(y,x)
}

Total_Residents <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs")
#____________________________________________________ these are to separate chunks
Num_Galago <- Dif_Years %>%
```

```r
  filter(Animal == "Galago")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Galago$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Galago$Year_Birth[i]:Num_Galago$Year_Death[i]))
  y = rbind(y,x)
}

Total_Galago <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Galago")
#____________________________________________________
Num_Lemur <- Dif_Years %>%
  filter(Animal == "Lemur")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Lemur$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Lemur$Year_Birth[i]:Num_Lemur$Year_Death[i]))
  y = rbind(y,x)
}

Total_Lemur <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Lemur")
#____________________________________________________
Num_Hybrid <- Dif_Years %>%
  filter(Animal == "Hybrid")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Hybrid$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Hybrid$Year_Birth[i]:Num_Hybrid$Year_Death[i]))
  y = rbind(y,x)
}

Total_Hybrid <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Hybrid")

#____________________________________________________
Num_Potto <- Dif_Years %>%
  filter(Animal == "Potto")

y = data.frame(yearpres=0)
```

```r
for( i in 1:length(Num_Potto$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Potto$Year_Birth[i]:Num_Potto$Year_Death[i]))
  y = rbind(y,x)
}

Total_Potto <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Potto")

#____________________________________________________
Num_Bushbaby <- Dif_Years %>%
  filter(Animal == "Bushbaby")


y = data.frame(yearpres=0)

for( i in 1:length(Num_Bushbaby$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Bushbaby$Year_Birth[i]:Num_Bushbaby$Year_Death[i]))
  y = rbind(y,x)
}

Total_Bushbaby <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Bushbaby")
#____________________________________________________
Num_Coquerel <- Dif_Years %>%
  filter(Animal == "Coquerel")

y = data.frame(yearpres=0)

for( i in 1:length(Num_Coquerel$Year_Birth)){
  x = data.frame(yearpres = rep(Num_Coquerel$Year_Birth[i]:Num_Coquerel$Year_Death[i]))
  y = rbind(y,x)
}

Total_Coquerel <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Coquerel")

#____________________________________________________
Num_Loris <- Dif_Years %>%
  filter(Animal == "Loris")


y = data.frame(yearpres=0)

for( i in 1:length(Num_Loris$Year_Birth)){
```

```r
  x = data.frame(yearpres = rep(Num_Loris$Year_Birth[i]:Num_Loris$Year_Death[i]))
  y = rbind(y,x)
}

Total_Loris <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Loris")

#_____________________________________________________
Num_AyeAye <- Dif_Years %>%
  filter(Animal == "Aye-Aye")


y = data.frame(yearpres=0)

for( i in 1:length(Num_AyeAye$Year_Birth)){
  x = data.frame(yearpres = rep(Num_AyeAye$Year_Birth[i]:Num_AyeAye$Year_Death[i]))
  y = rbind(y,x)
}

Total_AyeAye <- y %>%
  filter(y!= 0) %>%
  group_by(yearpres) %>%
  tally(name = "numberoflemurs") %>%
  mutate(Animal = "Aye-Aye")
#_____________________________________________________________________________

PrimatebyYear = rbind(Total_AyeAye,Total_Bushbaby, Total_Coquerel,Total_Galago,
                      Total_Hybrid, Total_Lemur, Total_Loris, Total_Potto)
```

```r
#This is the code for making two seperate graphs
#This is the plot for question 1
Q1 <- ggplot(PrimatebyYear, aes(x= yearpres, y = numberoflemurs, col = Animal)) +
  geom_line(size = 1) +
  scale_color_manual(values = c("#F0E442", "#E69F00", "#D55E00", "#009E73",
                                "#0072B2", "black", "#56B4E9", "#CC79A7"))+
  labs(x = "Year", y = "Total Number of Residents", title = "Panel A")+
  theme(legend.position = "none")

#This plot answers question 2
Q2 <- ggplot(Birth_duke, aes(x= Year_Birth, y = n, col = Animal)) +
  geom_point(size = 1, alpha = .7) +
  scale_color_manual(values = c("#F0E442", "#E69F00", "#D55E00", "#009E73",
                                "#0072B2", "black", "#56B4E9", "#CC79A7"))+
  labs(x = "Year", y = "Total Number of Babies", title = "Panel B")+
  theme(legend.position = "none")



Q3 <- grid.arrange(Q1, Q2, ncol = 2)
```

```r
plot1_legend <- ggplot(PrimatebyYear, aes(x= yearpres, y = numberoflemurs, col = Animal)) +
  geom_line(size = 1) +
  scale_color_manual(values = c("#F0E442", "#E69F00", "#D55E00", "#009E73",
                                "#0072B2", "black", "#56B4E9", "#CC79A7"))+
  labs(x = "Year", y = "Total Number of Residents", title = "Panel A")+
  theme(legend.position = "bottom")

get_only_legend <- function(plot) {

# get tabular interpretation of plot
plot_table <- ggplot_gtable(ggplot_build(plot))

# Mark only legend in plot
legend_plot <- which(sapply(plot_table$grobs, function(x) x$name) == "guide-box")

# extract legend
legend <- plot_table$grobs[[legend_plot]]

# return legend
return(legend)
}
legend <- get_only_legend(plot1_legend)

grid.arrange(Q3, legend, nrow = 2, heights = c(5, 1))
```

```r
grid.arrange(Q3, legend, nrow = 2, heights = c(5, 1))
```

```r
ggplot(mutated_born, aes(Year_Birth))+
  geom_histogram(col = "#0062B2", fill ="#0072B2")+
  labs(x = "Birth Year", y = "Number of Births")+
  xlim(1960,2020)+
  scale_x_continuous(breaks = seq(1960, 2020, by = 10))
```